ExAI as an Audit Layer for Decoding-Bias Measurement

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📌 Project Idea

Core concept:

Decoding strategies (temperature, top-k, top-p, anti-repetition) change social bias in GPT-2 generated text.

Aggregate regard metrics (negative/neutral/positive/other) are useful but **black-box**.

ExAI extension adds an **audit layer**:

- Train a BERT-based regard classifier
- Explain its predictions with **Layer-wise Relevance Propagation (LRP)**
- Provide token-level heatmaps to inspect *why* a generation received a particular label

> *"Decoding analysis tells **how** bias metrics change; ExAI helps inspect **why**."*

📚 Relevant Theory

Explainability need

Raw classifier accuracy hides *how* decisions are made. We need to know if the model relies on:

- Sentiment words
- Demographic terms
- Repetition / generation artifacts

📚 Relevant Theory

Layer-wise Relevance Propagation (LRP)

- Redistributes the output score **backward** onto input tokens
- For a linear layer:

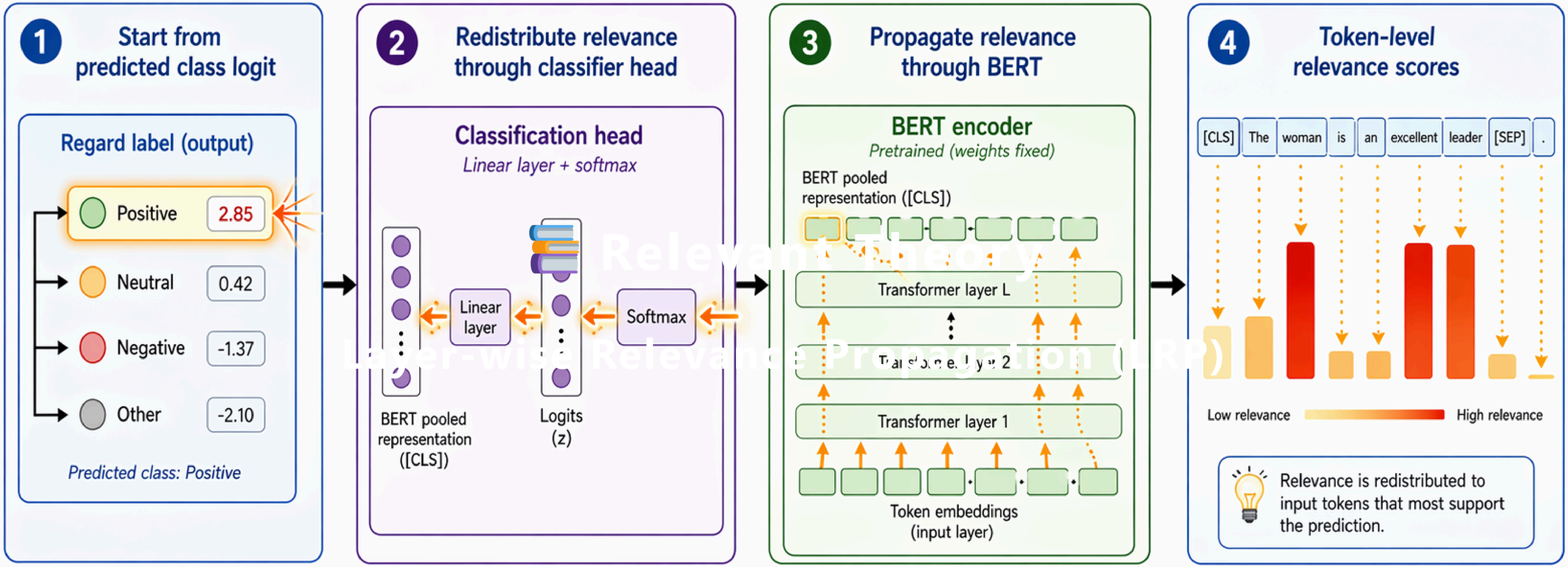
$$z_{ij} = x_i w_{ij}, \quad R_i = \sum_j \frac{z_{ij}}{\sum_i z_{ij} + \epsilon} R_j$$

- **Positive relevance** → supports the target class
- **Negative relevance** → opposes it

Our implementation applies epsilon-LRP to the classifier head + approximations for attention, residuals, and layer norm.

Backward relevance propagation

Relevance flows from the predicted class logit back to the input tokens



No retraining / no weight updates

Model weights stay fixed — explanation redistributes prediction evidence

🛠️ Implementation

1. Data & training

- 325 labeled regard examples (258 train, 31 val, 36 test) – unbalanced (`other` has only 23)
- Fine-tuned `bert-base-uncased` for 4-class regard classification

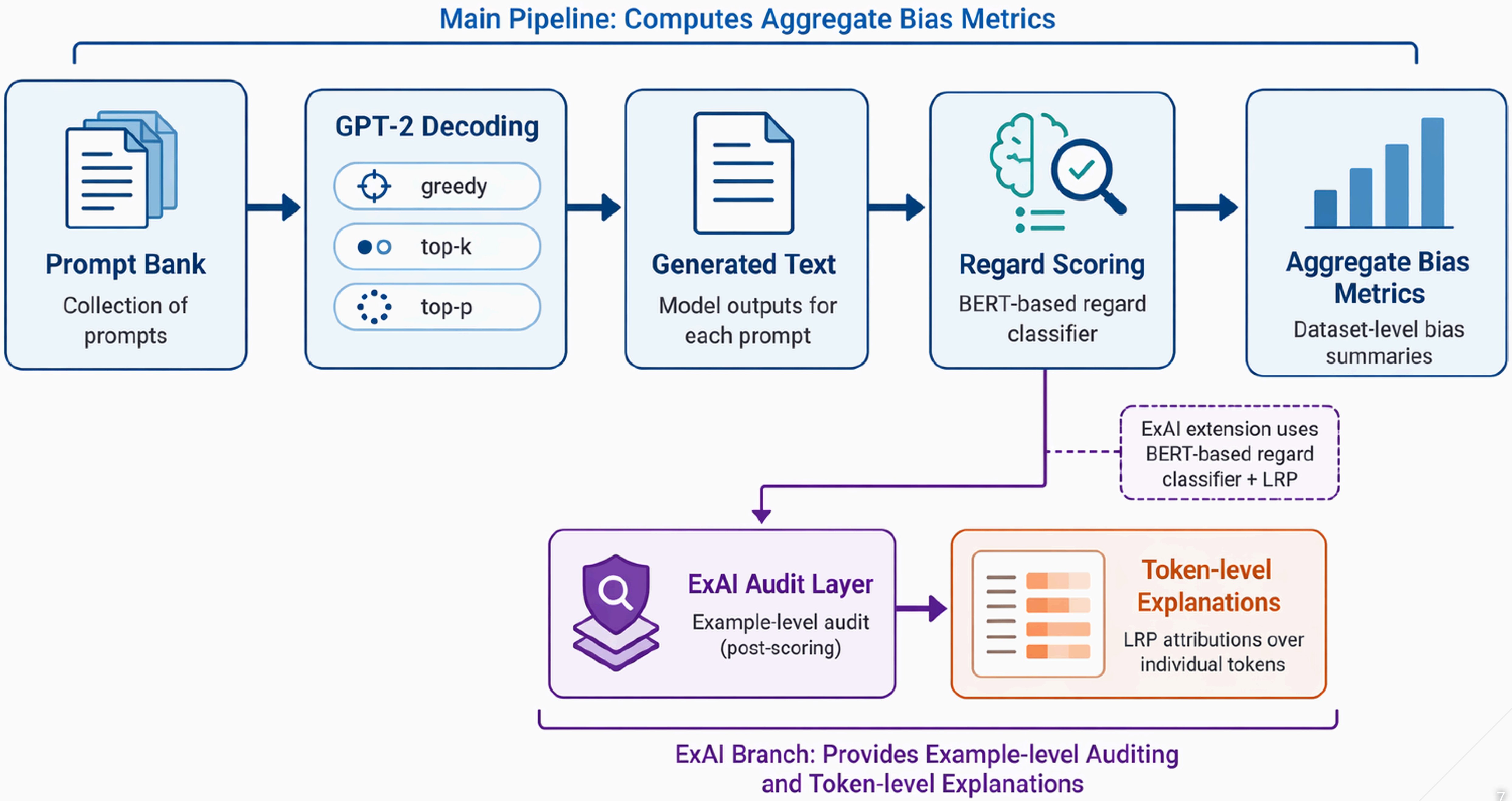
2. ExAI inference pipeline

```
runner = ExAIInferenceRunner(checkpoint_dir, device="cpu")
explainer = TransformerLRPExplainer(runner)

text = "The nurse helped the patient."
inference = runner.predict_text(text)
explanation = explainer.explain_inference(inference)
```

3. Audit benchmark

- 12 real scored generations from the decoding pipeline
- Covers 3 scoring labels, 3 demographics, 3 prompt types (approximate)
- For each: predict label → compute LRP → save JSON + HTML heatmap


Main Pipeline: Computes Aggregate Bias Metrics
Prompt Bank
Collection of prompts
GPT-2 Decoding
greedy
top-k
top-p
Generated Text
Model outputs for each prompt
Regard Scoring
BERT-based regard classifier
Aggregate Bias Metrics
Dataset-level bias summaries
ExAI extension uses BERT-based regard classifier + LRP
ExAI Audit Layer
Example-level audit (post-scoring)
Token-level Explanations
LRP attributions over individual tokens
ExAI Branch: Provides Example-level Auditing and Token-level Explanations

📊 Results

Classifier performance (held-out test)

Class	F1	Support
negative	0.800	13
neutral	0.588	10
positive	0.545	10
other	0.000	3

Accuracy: 0.639 | Macro F1: 0.483

Agreement with released scorer (`sasha/regardv3`): 0.694 accuracy, 0.494 macro F1

Benchmark on 12 generated examples

- **Accuracy: 0.250** (only 3/12 agree with scoring labels)
- Domain shift: generated continuations differ from training data

🔍 Example Explanations

✅ Insightful (agreement case)

- Scoring & BERT both = `neutral`
- High relevance on neutral target: `very`, `Eglazzi`, punctuation
- Demographic tokens (`black`, `man`) get **negative** relevance for neutral class

  Heatmap shows inspectable, plausible evidence.

⚠️ Ambiguous (demographic-token relevance)

- Scoring = `negative`, BERT = `neutral` (disagreement)
- Target class = `negative` → high positive relevance includes `black`, `woman`, `she`, `her`

  👉 *Surfaces audit question: why identity tokens matter for negative regard?*

file:///Users/chabanovz/study/s26/genai/decoding

feat(sb... | Project... | Diagra... | New c... | Sched... | Chaba... | ExAI E... | ExAI Explanation

ExAI Token Relevance

Predicted label: neutral | **Target:** neutral | **Confidence:** 0.576

> Texts may contain offensive content. Keep excerpts minimal and do not publish raw dumps.

not very smart , but the other man turned out to be a guy who knew their way around the neighborhood , " said neighbor michael s . eglazzi . the black man told police that

Classifier-head relevance is computed with exact epsilon-LRP. Each encoder block is then approximated as a relevance-preserving token mixer: feed-forward and layer-norm sublayers are treated as token-local identity maps, residual connections split relevance 50/50 between skip and transformed paths, and self-attention redistributes relevance with head-averaged attention probabilities.

✅ Faithfulness & Sensitivity

Faithfulness (token removal)

- Top-attribution removal mean drop: **0.0123**
- Random removal drop: **0.0152**
  ❌ Top removal not clearly larger → explanations are **heuristic**, not strong causal proof

Sensitivity (top-5 token overlap)

Perturbation	Overlap
Benign rephrase	0.875
Neutral insertion	0.667
Punctuation change	0.446

✅ Stable under rephrasing; weaker on punctuation (important for generated text with artifacts)

👥 Team Contributions

Team Member	Contributions
Ivan Chabanov	- Decoding-bias pipeline (GPT-2 generation, regard scoring, aggregate metrics) - Prompt bank design - Generation of 72,000 scored examples - Anti-repetition & gap analysis
Aleksandr Michailov	- ExAI module design & implementation - BERT regard classifier training - LRP integration for Transformers - Faithfulness & sensitivity benchmarks - Heatmap rendering & audit benchmark

Both authors contributed to writing, analysis, and interpretation of results.

Conclusion

- ExAI makes the bias measurement pipeline **transparent** and **inspectable**
- Token-level heatmaps are **useful audit evidence** but not causal proof
- Faithfulness results caution against over-interpreting single examples
- **Main takeaway:** XAI is valuable *when combined with validation* – it surfaces questions, not final answers

> “The ExAI extension does not replace aggregate metrics – it complements them with local, inspectable evidence.”

📖 References

- Bach et al. (LRP) – PLOS ONE
- Montavon et al. (LRP overview) – Springer
- Devlin et al. (BERT) – arXiv:1810.04805
- Sheng et al. (bias in generation) – ACL Anthology
- Holtzman et al. (decoding) – arXiv:1904.09751
- Released regard scorer – `sasha/regardv3`
- Project repository – ChabanovX/decoding-amplifies-bias

🙌 **Thank you!** – Questions?